Increasing electricity exports through delivery-informed allocation in distribution networks

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Abstract

Export access can be divided among distribution-network users before they know how much electricity they can supply. An unused award might then block another user, but the shortfall alone cannot show whether delivery was lost. We link requests, network-constrained awards and uncertain output to identify a case in which a larger request benefits its sender while reducing total delivery. We then introduce externality-conditioned counterfactual feedback (ECF). After an interval, it reruns the allocation as if users with unused awards had requested only what they delivered. Network conditions and the allocation rule stay fixed. Later request ceilings fall only if the alternative passes AC power-flow checks and increases delivery elsewhere and in total; otherwise they recover. In controlled medium-voltage tests across five allocation rules, ECF delivers 24.8 kW more on average than a control that reduces the same total request across all participants. In an 80-day public-data-grounded Australian network simulation, ECF delivers 1.39 MWh more modeled electricity than no feedback. Across 12 replay-free schedules with similar update frequencies, ECF delivers 1.34 MWh more on average. The benefit depends on identifying when unused access can serve electricity available elsewhere, not merely on reducing requests.

1 Introduction

Distributed generation, including rooftop photovoltaics (PV), can supply the wider power system when local networks can carry electricity not consumed on-site. Voltage and equipment limits restrict simultaneous export at some locations and times. Operating envelopes translate those limits into connection-level bounds[1–3] and are entering flexible export arrangements in several jurisdictions[4–7]. Where several connections share a binding limit, access granted to one leaves less for others. Allocation therefore determines how much available electricity reaches the network, not just who has permission to export.

The difficulty is that requests arrive before available output is known. A large request can protect a participant's export if generation is high, yet leave part of its award unused if generation is low. This can be a reasonable response to uncertainty rather than an attempt to withhold access[8–10]. The unused award matters to the rest of the network only if another connection could have delivered more with a different allocation.

Fig. 1 separates these possibilities. Connections A and B share a binding limit. In (b) and (c), B leaves the same amount of access unused. Reallocating it does nothing in (b) because A has no additional electricity available. In (c), A could use the released access, so total delivery would rise.

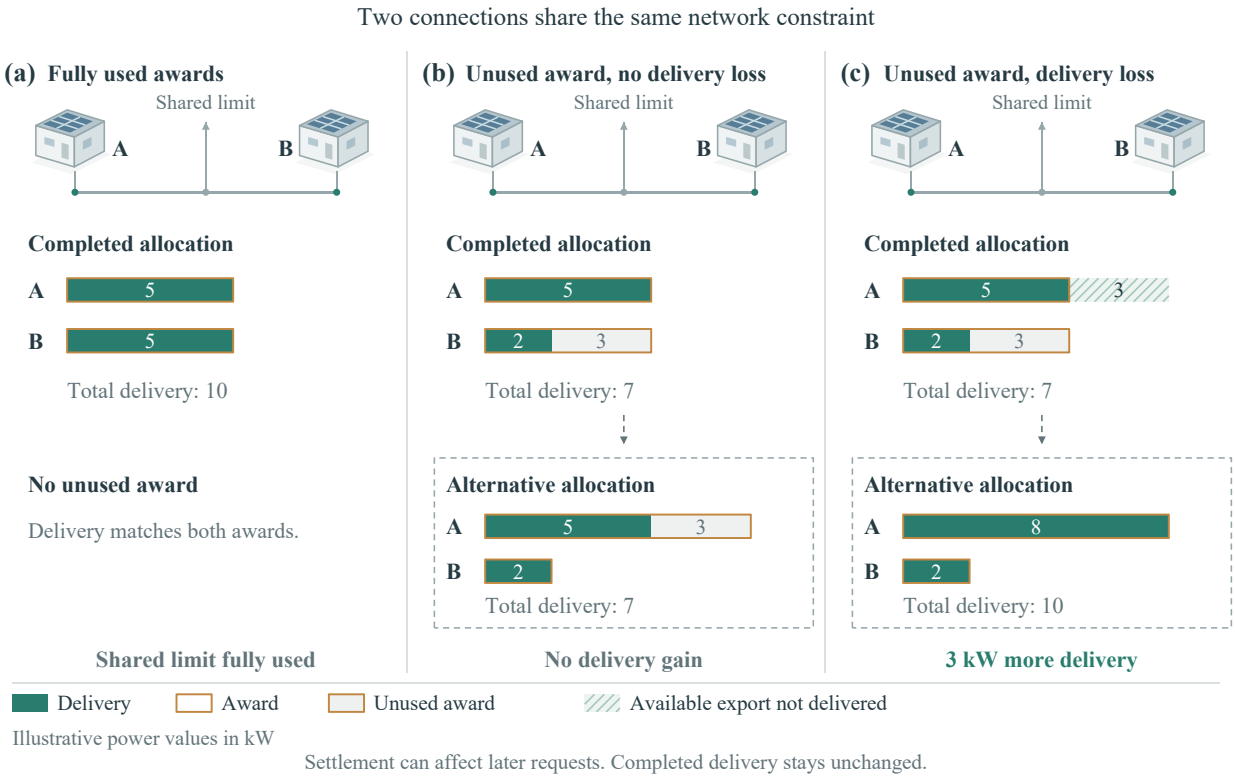


Figure 1: When unused export permission displaces delivery. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW authorization bound and initial awards of 5 kW each; they are not case-study results.

B's award and metered delivery look alike in the two cases. They reveal unused permission, but not its cost to another exporter.

A shortfall charge can discourage over-requesting, but B's meter reading cannot tell whether A lost delivery. Reducing everyone's request can also avoid idle awards while denying access to participants whose output is high. The useful distinction is between an award that went unused and one whose release could have increased delivery elsewhere. Completed operating records can inform that distinction and guide the next allocation[11,12].

Existing approaches address other parts of this decision. Operating-envelope methods calculate export limits from voltage and equipment constraints, including uncertain feeder conditions[1,2,13,14]. Flexible-limit programmes describe how those limits can be issued and exchanged[15–17]. Allocation rules divide access by proportional, equality or bargaining objectives[18–25]. Local measurements, past curtailment and persistent allocation states can also guide future limits[26–28]. Network-checked exchange can transfer an issued limit before delivery[29]. Each approach addresses a particular stage

of access management. We study the decision after delivery: whether reallocating unused access in that completed interval could have increased export elsewhere.

This question cannot be answered by treating every large request as harmful. Energy-sharing and flexibility studies show why bids, prices and uncertainty affect requests[8,30–35]; reported-baseline studies show that private incentives can change reports[9,10,36]. A useful response must distinguish a request that preserves valuable export from one that denies it elsewhere.

We first use one uncertain-output model for both request choice and later delivery. It reveals cases in which a larger request raises the sender's expected payoff and unused access while lowering total delivery. We call this a request–access–delivery externality. The model also produces productive large requests, for which total delivery does not fall. These two outcomes make a blanket response to high requests unsuitable.

We then test a response based on completed delivery. The proposed externality-conditioned counterfactual feedback (ECF) asks whether reducing the unused-award group's requests in a completed interval would have released access that others could actually use. Only a feasible increase in outside and total delivery lowers that group's later request ceilings. The completed award remains untouched, and the original allocation rule remains in place. The test identifies an opportunity for the group as a whole; it does not assign responsibility to every member.

Fig. 2 shows the sequence. Once an interval ends, ECF repeats the allocation with unused-award requests reduced to their completed delivery. It holds the exogenous network conditions and allocation rule fixed. A post-event estimate of available export then tests whether the alternative could have increased delivery elsewhere and in total without failing the specified AC checks. This estimate must remain valid under the alternative award. A qualifying result lowers the next request ceiling; otherwise the ceiling recovers. The controlled study uses simulated available export; the Australian study constructs an estimate from public PV and load profiles and therefore reports modeled, not metered, delivery.

The experiments follow the two choices ECF makes. Controlled medium-voltage cases first establish the request conflict and test whether reducing requests at the affected connections preserves more delivery than a reduction of the same total amount elsewhere. The Australian simulation then tests whether a completed-state replay selects more useful update intervals than controls that update without it. Output histories and electrical placement test how far those results extend.

Table 1 distinguishes the comparison mechanisms. The equal-total control tests where requests are reduced. The unqualified and frequency-calibrated random updates test what is gained by selecting update intervals through replay. Other controls change request incentives, use a participant's delivery ratio, or omit the forecast increment.

2 Results

The results first test whether a request that helps its sender can reduce delivery elsewhere. We then ask whether a targeted reduction preserves more useful export than an equal-total reduction, and whether completed-state replay selects better update intervals than a similar number of replay-free updates. The controlled study reports mean power in kW; the Australian simulation reports modeled energy over the assessed days. Differences are calculated before rounding and name their comparators.

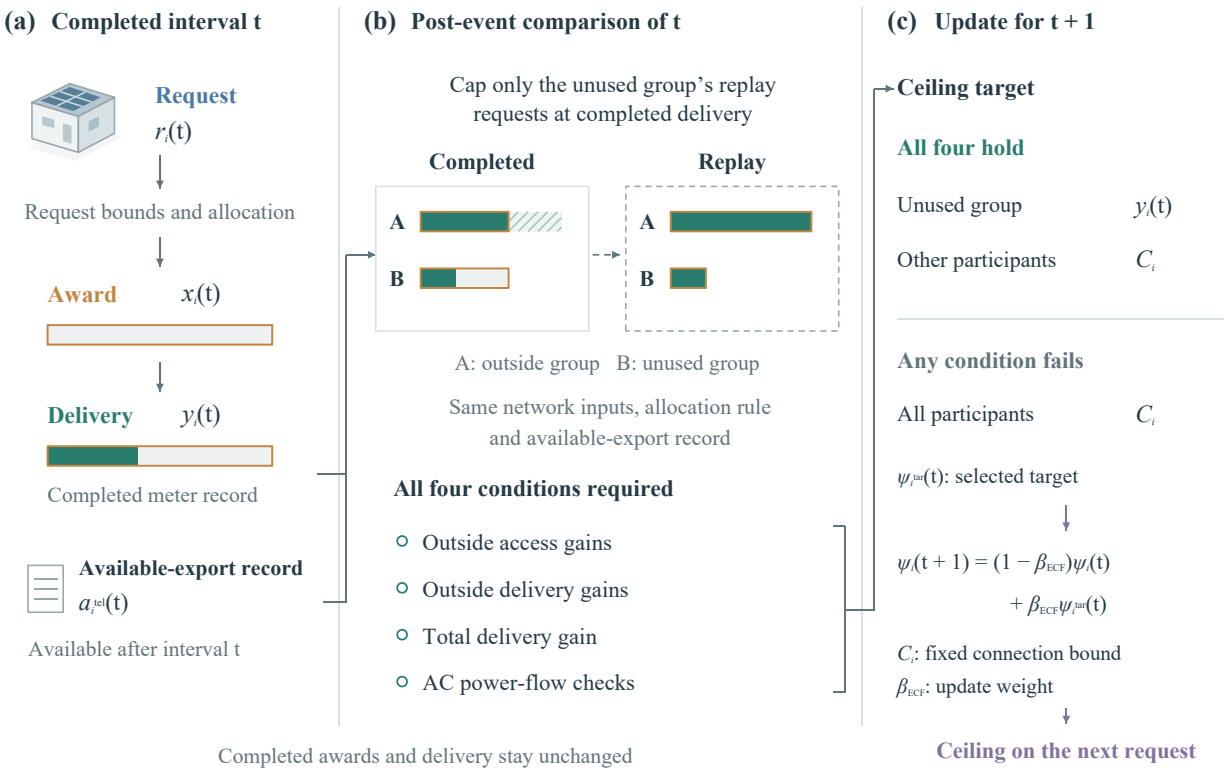


Figure 2: Using completed delivery to update later request ceilings. (a) Submitted requests pass through connection and current feedback bounds before the existing rule allocates access. (b) After delivery, requests from the unused-award group are reduced to completed delivery and allocation is repeated under the same exogenous network conditions. The comparison uses a post-event available-export record and checks whether delivery could rise outside the group and in total without violating the specified AC limits. (c) A qualifying comparison sets a lower target for the group's next request ceiling; otherwise, ceilings recover toward connection bounds. The completed award and delivery do not change.

2.1 Private delivery gains can accompany a larger loss elsewhere

In 1,500 controlled parameter–location cases without a shortfall cost, a larger request raises the requester's expected delivery by 5.2 kW on average. Other participants lose 18.8 kW, leaving total delivery 13.5 kW lower. The requester values its own gain but does not bear the larger loss elsewhere. This is the request conflict that motivates feedback. Request choice and delivery come from the same uncertain-output model, so the contrast follows from how the network reallocates access.

Fig. 3 traces the request choice as forecast uncertainty σ and normalized shortfall cost ϑ change. The cost parameter is the modeled charge for one unit of unused award divided by the value of one unit delivered. We test five allocation rules—weighted proportional (WP), equal reduction (ER), equal access (EA), maximum export (ME), and fairness-aware allocation (FA)—under two reactive-power settings. Each larger request is compared with the same participant's forecast-based request; Methods specifies the output model.

The mean loss hides two different kinds of larger request. We classify a request–access–delivery externality when the requester's expected payoff and unused award rise but total delivery falls. A

Table 1: How the comparison mechanisms differ. C denotes the controlled study and A the Australian simulation. Each paired comparison uses the same network and available-export trajectory.

Mechanism	What changes	Question tested
No feedback (C, A)	Ceiling stays at the connection bound.	Baseline delivery.
Shortfall cost (C)	A modeled cost changes the initial request.	Effect of an incentive before allocation.
Mismatch-only feedback (C)	A participant's own delivery-to-award ratio updates its ceiling.	Effect of responding to shortfall alone.
Update without replay test (A)	ECF's formula applies after every unused award.	Effect of removing qualification, including more frequent updates.
Frequency-calibrated random update (A)	The same formula updates randomly selected unused-award intervals.	Whether a similar update frequency explains ECF's gain.
Equal-total reduction (C, A)	ECF's total reduction is spread proportionally.	Value of where requests are reduced.
Previous-day persistence (A)	Request uses the previous-day output estimate alone.	Cost of a conservative request.
ECF (C, A)	Qualified unused awards lower later ceilings.	Combined effect of testing and updating.

productive large request raises payoff without a material total-delivery loss. In the first case, the requester delivers more while the others lose still more; the implication and numerical classification are given in Methods and Supplementary Section S2.

At $\vartheta = 0$, 71.5% of the 1,500 cases meet the externality definition, while 11.9% are productive large requests. The model selects the reference request in 14.7%; the remaining 1.9% meet neither large-request definition. These classes coexist at the same cost setting. Request size or unused access alone cannot reveal which outcome occurred.

Uncertainty makes the higher-output outcome worth protecting; a higher shortfall cost instead discourages unused awards. At $\vartheta = 10$, 10.7% of cases still meet the externality definition, but the mean total-delivery loss falls to 0.6 kW when zero-loss cases are included. Productive large requests still account for 11.5%. A charge can suppress the conflict, but it also changes requests that would have been useful.

Panel (d) tests whether one shortfall price could reflect the loss to others. It measures that loss per additional unit of unused award for each 0.1 step in the request multiplier. Across 4,608 steps that raise own delivery and unused access while lowering total delivery, the median diagnostic ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$ is 2.29, with an interquartile range of 1.10–5.72. Rule-specific medians run from 0.92 for ME to 5.37 for EA. The spread means that no single cost matches the modeled delivery loss at every tested location and state. Supplementary Section S2 derives the ratio and its domain. We next ask whether feedback can respond to the delivery opportunity revealed after an interval, without suppressing every large request beforehand.

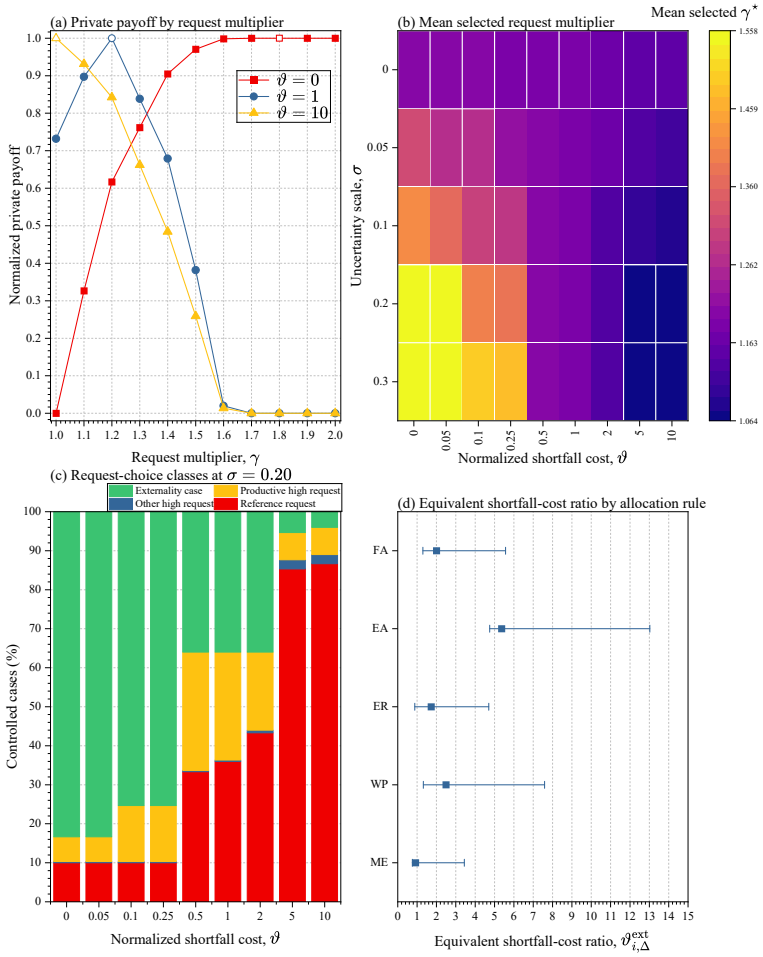


Figure 3: How a larger request becomes productive or harmful before feedback. (a) Representative private-payoff curves for three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. The mean covers participants, allocation rules, and reactive-power settings. (c) Shares at $\sigma = 0.20$ ($n = 300$): reference requests; productive high requests, which raise payoff without lowering total delivery; request–access–delivery externality cases, which raise payoff and unused awards while lowering total delivery; and other high requests, which meet neither high-request definition. (d) Finite-difference cost ratio $\vartheta^{\text{ext}}_{i,\Delta}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

2.2 Targeted request reductions preserve useful export access

We next follow requests through 16-interval output trajectories. Feedback cannot change the award that produced a shortfall; it changes only later requests. At zero shortfall cost and an intermediate feedback gain, $\beta_{\text{ECF}} = 0.5$, ECF delivers 24.8 kW more on average than a control that reduces the same total request across all participants. The difference must therefore come from where the reductions fall. A second comparison asks whether a charge before allocation, or feedback based

only on one's own delivery, preserves as much useful export. All controls start from the same raw requests and network conditions. Supplementary Fig. 1 illustrates how completed-interval evidence affects the next interval as available output falls and recovers.

2.2.1 A shortfall cost reduces displacement but also useful delivery

A shortfall cost changes requests before anyone receives an award. Raising ϑ from 0 to 1 reduces the mean multiplier from 1.51 to 1.14 and the share of intervals meeting the economic externality definition from 50.1% to 15.6%. But it also removes 45.6% of the larger-request group's delivery gain (Supplementary Fig. 2). At $\vartheta = 10$, none of these intervals meets the externality definition, while the group's gain is almost zero. The charge reduces the modeled conflict partly by removing useful high-output access as well. This economic classification is not the replay test that activates ECF.

At $\vartheta = 0$, the no-feedback case is the common baseline. Relative to forecast-based requests, its mean total delivery is 21.7 kW lower. ECF instead raises total delivery by 3.2 kW, a 24.9-kW improvement over no feedback, while retaining 79.4% of the larger-request group's original delivery gain. Feedback based only on each participant's delivery-to-award ratio recovers 15.2 kW relative to no feedback, leaving ECF 9.7 kW ahead. Supplementary Section S3 reports the group decomposition and other cost settings.

2.2.2 Where requests are reduced determines recovered delivery

The equal-total control sharpens this comparison. It removes exactly the same total request as ECF in each interval, but spreads the reduction among participants. ECF's 24.8-kW mean advantage at $\vartheta = 0$ therefore isolates the value of placing the reduction. ECF beats this control in 95.6% of 90 matched trajectories; it beats delivery-ratio feedback in 73.3%. Its mean advantage over both remains positive at $\vartheta = 1$ and 10.

Selecting a useful interval is not sufficient if the next ceiling is too tight. At $\vartheta = 0$, ECF changes mean total delivery relative to forecast-based requests by -2.8, 3.2 and -23.4 kW as its update gain β_{ECF} rises from 0.25 to 0.5 and then 1 (Fig. 4). The reversal also occurs at the other shortfall-cost settings. At unit gain, one completed interval replaces the previous ceiling; output may then recover before the ceiling does. The intermediate gain is the primary comparison setting. It gives the largest mean delivery of the three tested gains, but the sweep does not establish a universal optimum.

In the controlled experiment, 12,910 of 12,960 replay attempts meet the four conditions, with a median 37.8-kW potential recovery in the completed interval. Yet an unused award alone is insufficient: 20 of 4,320 attempts at the primary gain fail because total delivery would not rise. The public-data simulation later provides a more variable setting in which only 12.6% of intervals qualify.

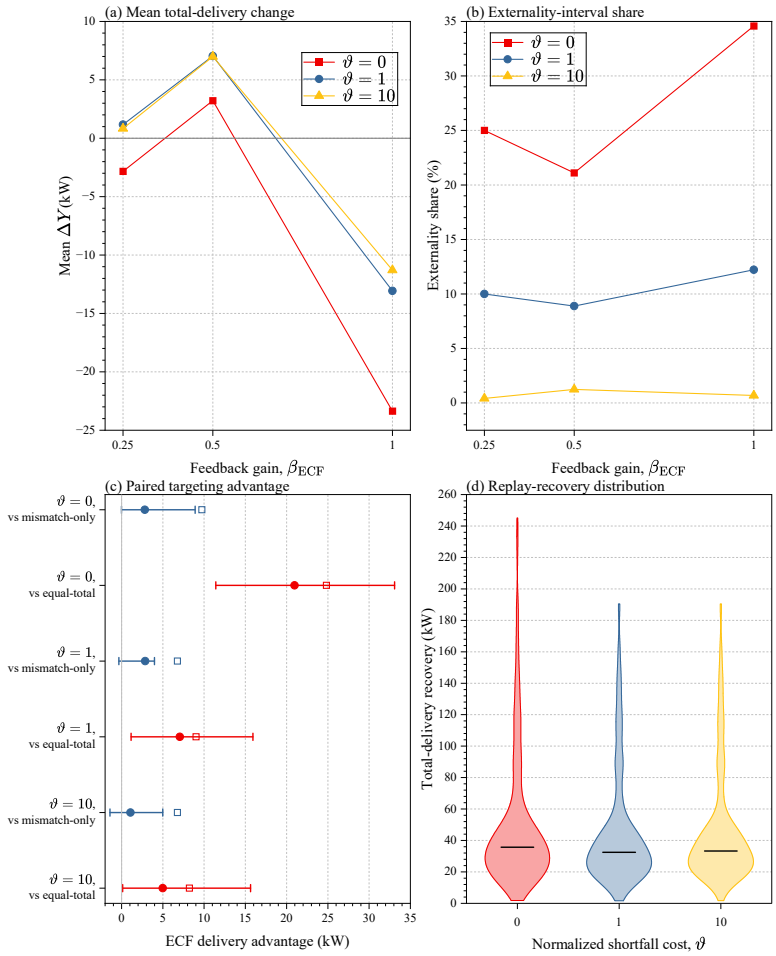


Figure 4: Targeted feedback preserves delivery, but a unit-gain update reverses the improvement in the controlled cases. (a) Mean signed change in total delivery and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labelled "vs mismatch-only" compare ECF with mismatch-only feedback. Rows labelled "vs equal-total" compare ECF with the equal-total control. Panel (c) uses the primary gain, $\beta_{\text{ECF}} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

2.3 Output timing and allocation rules alter the delivery gain

The preceding comparison could still be peculiar to one allocation rule or one ordering of output. Fig. 5 varies both, along with the location of changed requests and reactive-power treatment. At $\vartheta = 0$ and $\beta_{\text{ECF}} = 0.5$, mean ECF delivery is higher than both no feedback and equal-total reduction at every tested factor level. The advantage over no feedback ranges from 8.9 kW under FA to 67.5 kW under EA; against equal-total reduction it ranges from 9.7 to 50.8 kW. The effect is not specific to one of the five rules, although its size depends on the rule.

The three temporal profiles use the same available-export values with the same frequencies; only their order changes. ECF gains 35.3 kW over no feedback when low-output intervals are clustered,

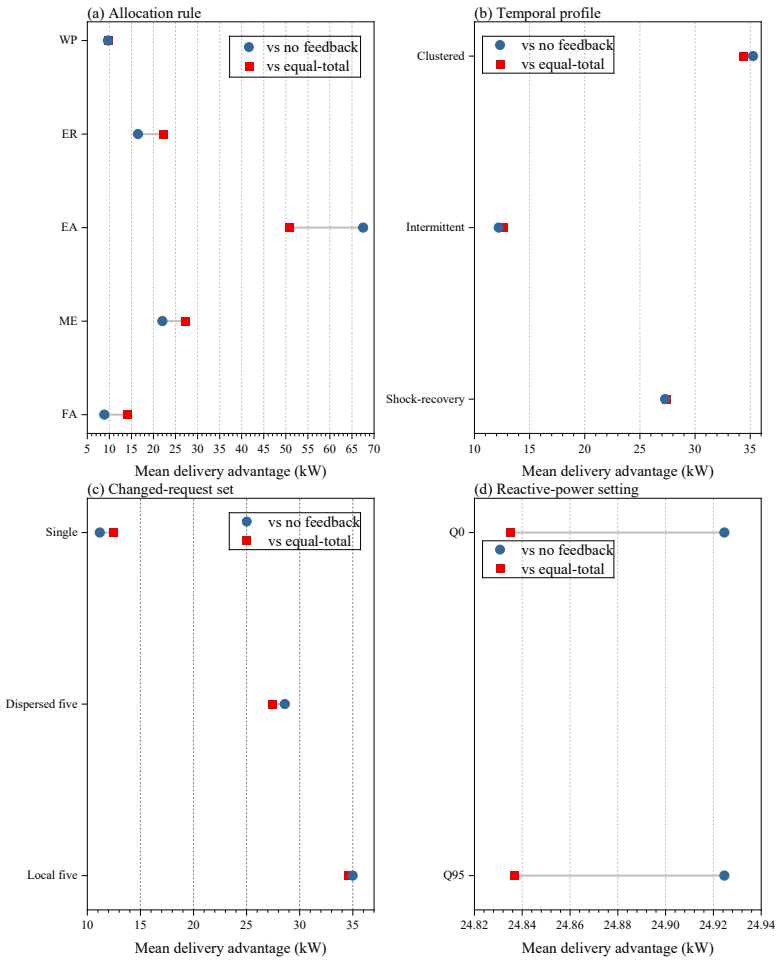


Figure 5: The controlled delivery advantage persists across allocation rules and output timing at $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$. Results are grouped by (a) allocation rule, (b) temporal profile, (c) changed-request set, and (d) reactive-power setting. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption.

12.2 kW when they are intermittent, and 27.3 kW in a shock–recovery sequence. The respective gains over equal-total reduction are 34.4, 12.6 and 27.4 kW. Thus an update based on a completed low-output interval has a different effect when similar states persist than when output changes rapidly. The result concerns timing, not a different set of available-output values.

When one participant changes its request, the mean gain over no feedback is 11.2 kW. It is 35.0 kW for five nearby participants and 28.6 kW for five dispersed participants. Each chooses its own request multiplier. The latter two cases change participant identity as well as electrical placement, so their difference cannot be attributed to distance alone. Zero participant reactive power (Q0) and 0.95-power-factor absorption (Q95) both give a 24.9-kW mean gain over no feedback at the reported precision, and 24.8 kW over equal-total reduction.

2.3.1 AC feasibility and allocation equality

Delivery recovered by the model must also be physically deliverable. Both sides of all 25,920 matched award comparisons pass nonlinear AC checks for convergence, power balance, voltage and apparent power at both branch ends. Awards can exceed actual output, so we check delivery injections separately: all 22,520 distinct observed- and replay-delivery states pass the same checks. Maximum branch utilization is 90.91% of the fixed synthetic rating. Supplementary Fig. 3 and Section S3.5 give voltage margins and state counts. These checks verify the declared controlled operating envelope, rather than native equipment ratings for the published feeder.

More delivery need not mean more equal access. The capacity-normalized Jain index in Supplementary Fig. 3 measures equality of awards, not social fairness. At $\vartheta = 0$, ECF improves its mean change relative to no feedback under all five rules, but four remain below the forecast-based reference. EA moves from -2.4 percentage points without feedback to 0.1 with ECF; WP, ER, ME and FA remain between approximately -0.8 and -0.5 points. ECF also exceeds equal-total reduction on this mean index for every rule. The additional authorization and unused-award results appear in the Supplementary Information.

2.4 Public-data replay gains delivery through a different allocation of access

2.4.1 Better use of access increases delivery, not just utilization

We next test the targeting contrast with a published Australian medium- and low-voltage network and public household PV and load histories. Across 80 assessed network-days, ECF delivers 239.36 MWh of modeled export: 1.39 MWh (0.59%) more than either no feedback or equal-total request reduction (Fig. 6). Requests and available export are constructed from those histories; they are not linked operational bids or metered curtailed export. The primary comparison uses proportional allocation and initializes ceilings each day.

The gain is not produced by shrinking the total award. Relative to no feedback, ECF authorizes 1.206 MWh more and leaves 0.186 MWh less unused. The two changes add to the 1.392-MWh delivery gain by the accounting identity in (8). Equal-total reduction has almost no delivery effect because proportional scaling preserves request shares: a binding network limit can be met by changing the common allocation factor instead. Targeted feedback changes those shares. Supplementary Sections S4 and S11.1 give the derivation and full energy balance. The identity describes the outcome; it does not separately attribute causality to either term.

The gain is not uniform across days. ECF delivers more on 44, less on 32 and the same amount on four, relative to either control. Its mean gain is 17.4 kWh per assessed day; the dependence-adjusted 95% intervals are 5.1–29.7 kWh against no feedback and 4.9–29.9 kWh against equal-total reduction. A continuous 35-day block contributes 1.06 MWh of the 1.39-MWh gain. The other 45 days contribute 0.33 MWh, with an interval that includes zero when considered alone (Supplementary Section S10.3). ECF qualifies 483 of 3,840 half-hours (12.6%), so most completed shortfalls do not tighten the next request ceiling.

The added export is visible at the feeder boundary. Net source exchange rises by 1.377 MWh relative to either control, about 98.9% of the connection-point gain. Because the feeder remains a net importer, this is a reduction in upstream import; Supplementary Section S11.1 gives the circuit loss and passive-element terms. By contrast, previous-day persistence uses 89.0% of its authorization but delivers 50.33 MWh less than ECF. It omits the forecast increment and therefore requests too

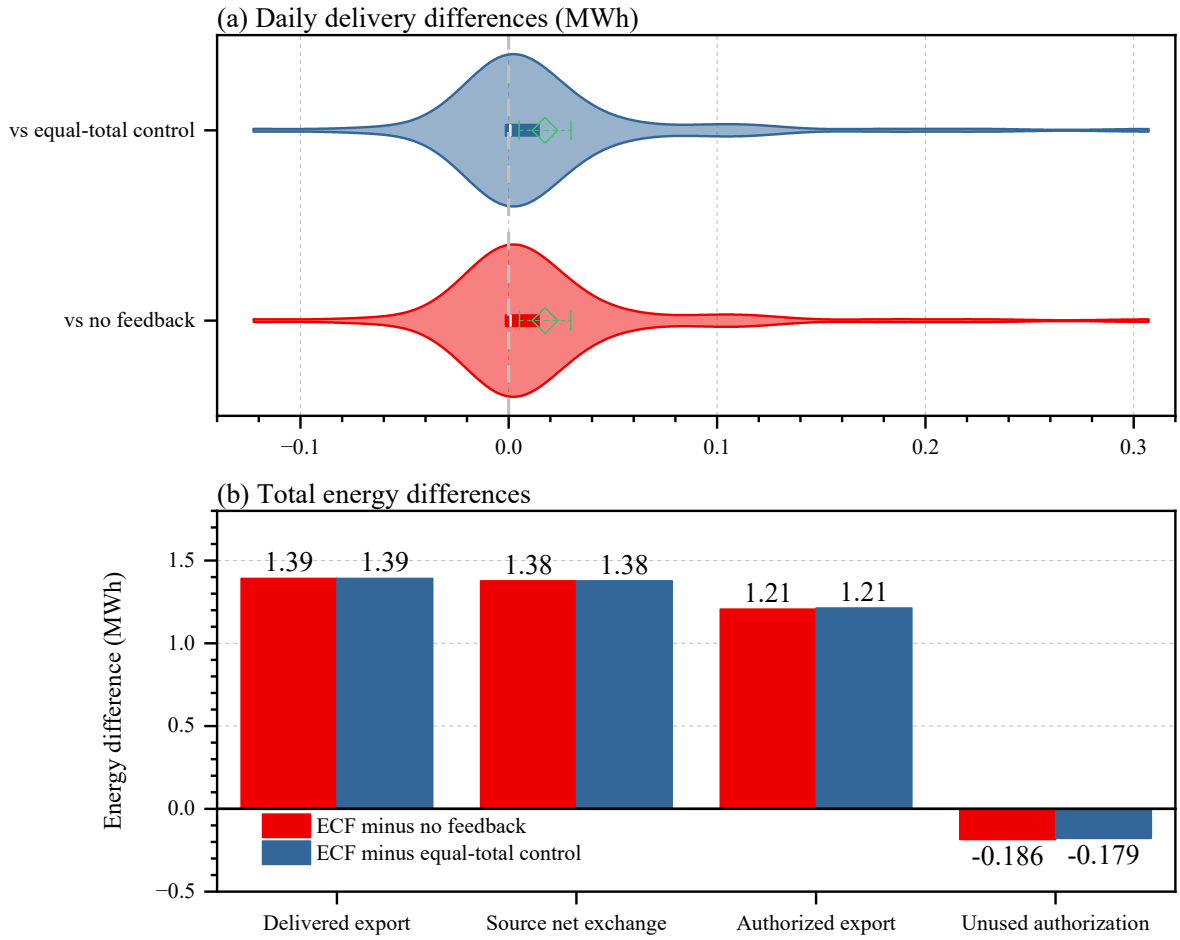


Figure 6: Selective feedback increases modeled delivery through more authorization and less unused access in the Australian public-data replay. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

little on many high-output intervals. Utilization alone rewards a small, fully used award even when more electricity could have been delivered.

All primary trajectories pass the specified allocation and delivery screens. Network limits reduce the proportional allocation factor in 44.4% of ECF half-hours; the common network constraints remain active even as the distribution of requests changes.

Table 2: Modeled export over the same 80 Australian network days. The unqualified update uses ECF's ceiling formula and gain after every unused award. The frequency-calibrated random row averages 12 replay-free schedules with the same formula and gain. The equal-total control matches ECF's request reduction in each half-hour. All energy totals are in MWh and are rounded independently.

Control	Authorized	Unused	Delivered
No feedback	328.490	90.526	237.963
Equal-total reduction	328.483	90.519	237.964
Update without replay test	305.159	72.226	232.933
Frequency-calibrated random update	328.357	90.345	238.012
ECF	329.696	90.341	239.356

2.4.2 Replay qualification avoids overrestricting later requests

Only 12.6% of intervals qualify, so the next question is whether the replay adds information or merely reduces the number of updates. We first remove the replay test but keep the same ceiling formula and gain. Updating after every unused award lowers modeled delivery from 239.36 to 232.93 MWh. ECF delivers 6.42 MWh (2.76%) more and wins on 75 of 80 days. The unqualified rule leaves 18.11 MWh less authorization unused, yet grants 24.54 MWh less authorization. It removes useful export along with idle awards.

That comparison also changes update frequency: 3,086 half-hours rather than ECF's 483. A second control keeps the target and gain but selects unused-award intervals without replay. Its 12 fixed schedules update in 463–519 half-hours. Every schedule delivers less than ECF, by 0.92–1.60 MWh. On average they deliver only 0.05 MWh more than no feedback, versus ECF's 1.39 MWh gain. They also leave almost the same authorization unused as ECF (90.345 versus 90.341 MWh). After averaging schedules within each date, ECF gains 1.34 MWh, or 16.8 kWh per assessed day. It wins on 66 of 80 days, with a dependence-adjusted 95% interval of 7.8–25.8 kWh per day. Selecting the update interval matters beyond updating less often in this comparison. Supplementary Sections S11.2–S11.3 give the full trajectories and design.

An informed choice of interval still needs a suitable update size. At $\beta_{\mathrm{ECF}} = 0.25$, ECF gains 1.34 MWh over each of no feedback and equal-total reduction. At unit gain, it loses 3.04 MWh relative to no feedback: replacing the ceiling with one interval's target can deny export after output recovers. This adverse result remains when ceilings carry across midnight. In ten consecutive-date blocks covering the 80 assessed days, total delivery at each tested gain differs from daily initialization by less than 0.1 kWh (Supplementary Section S12.2).

Nor does the advantage require participants to repeat fixed requests indefinitely. When they adjust requests using their own delivery history, ECF delivers 1.44 MWh more than no feedback and 4.53 MWh more than the same update without qualification. Each control develops its own request history. This experiment tests one form of adaptation, not strategic anticipation of ECF; Supplementary Section S14.2 specifies the model.

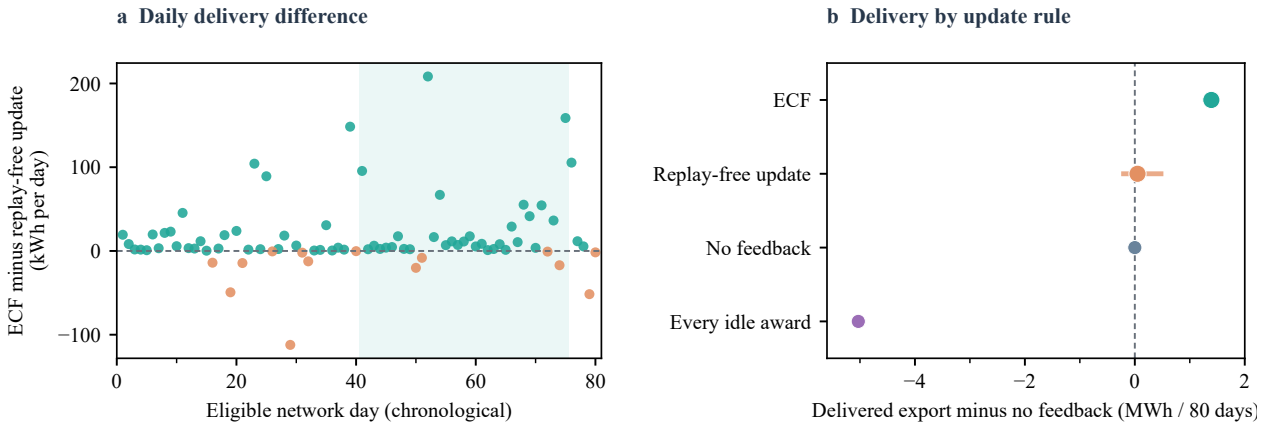


Figure 7: Replay-selected updates recover more modeled delivery than comparably frequent replay-free updates in the Australian simulation. (a) Paired daily modeled-delivery difference between ECF and the mean of 12 frequency-calibrated random updates. Positive points favor ECF; shading marks the longest continuous 35-day block. (b) Eighty-day delivery relative to no feedback. The central random marker is the mean across 12 fixed schedules; its horizontal range shows their individual totals. The number of updated half-hours is 483 for ECF, 463–519 for the random schedules, 0 for no feedback and 3,086 for updating after every unused award. The schedules match ECF's update frequency without using replay to choose their intervals.

2.4.3 Record error changes which intervals qualify

The replay also depends on its estimate of available export. We scale that record by ±10%, delay it by one interval, or substitute modeled delivery for 10% of records. ECF retains a positive mean advantage over both controls in these tests (Fig. 8): 16.3–18.0 kWh per assessed day, with dependence-adjusted 95% intervals above zero. This mean result does not ensure that every individual replay decision is correct.

When a validated error bound is available, using its conservative lower limit prevents false qualifications in the tested overestimation cases. The tradeoff is missed delivery: at the largest tested error radius, 258 of 496 feasible recoveries are missed and total delivery is 269 kWh below the point-record case. Both approaches still improve on no feedback in the assessed trajectories (Supplementary Section S12.3). A conservative replay protects the decision, but can forgo an opportunity.

Fig. 8 collects the gain and record responses, together with the complete-customer cohort. The latter broadens the calendar to 361 days at the same load multiplier of 0.30. ECF gains 10.96 MWh over each of no feedback and equal-total reduction, or 30.4 kWh per network day; both mean-difference intervals remain above zero and the specified physical checks pass. Customer composition and calendar coverage change together, so this comparison does not isolate either factor (Supplementary Section S10.4).

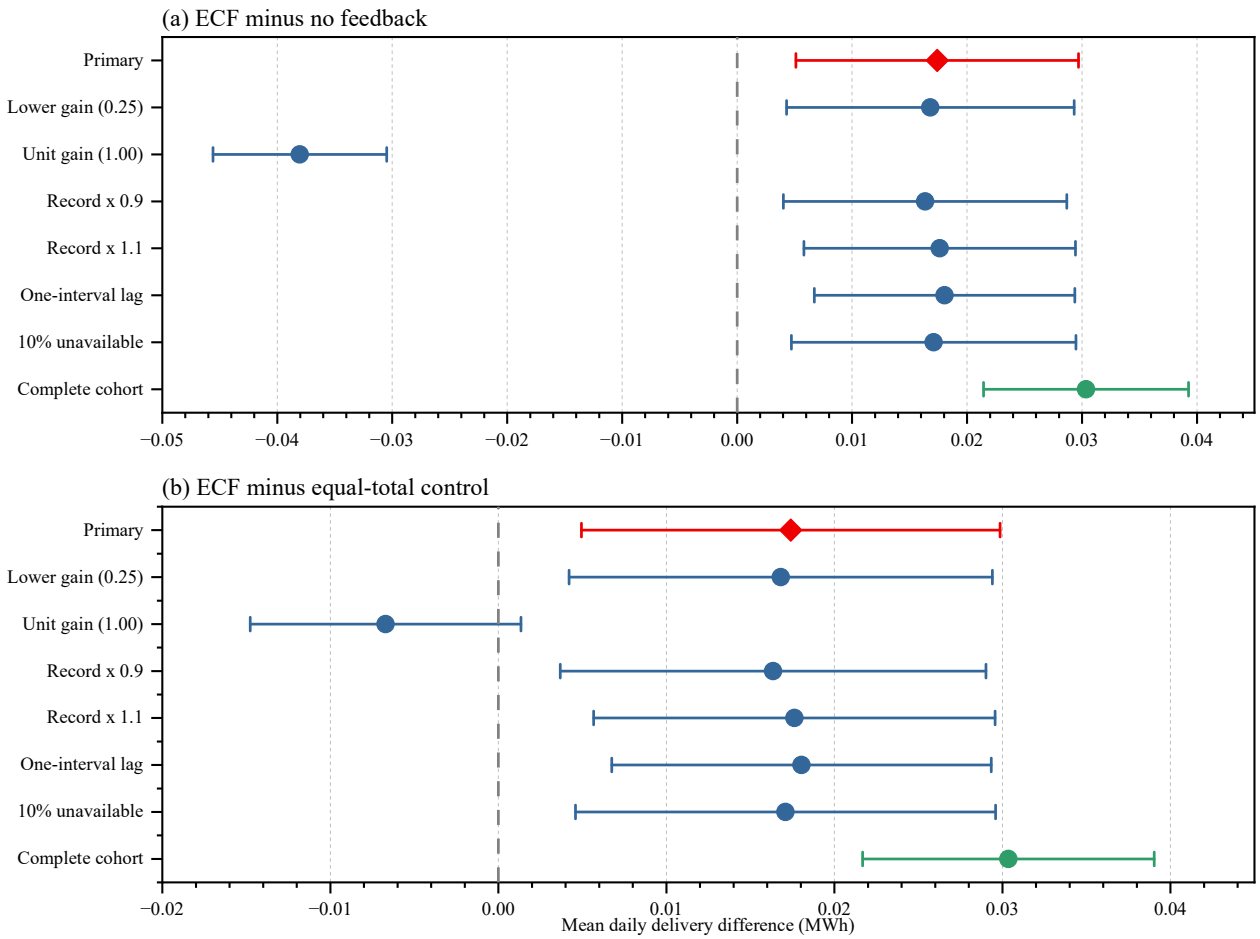


Figure 8: Gain, record quality and customer coverage change the Australian delivery result. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage. Every transformed available-export record is floored at modeled delivery. All settings pass the specified allocation and delivery checks; the zero line separates positive from negative mean delivery differences.

2.4.4 Spatial placement and available generation shape the benefit

The primary profile-to-service mapping is constructed because the public household and network sources have no common customer identifier. We therefore repeat the 80-day comparison with five alternative placements of the same profiles. ECF gains about 0.9–5.1 MWh over either control, with positive pointwise mean-difference intervals in all five. At five PV-capacity levels from half to one-and-a-half times the primary level, the gain is 1.39–1.88 MWh. Placement and available generation change the size of the effect, but neither reverses its sign in these tests. The gain does not scale monotonically with installed PV (Supplementary Sections S13.1–S13.2).

Changing the network itself reveals both gains and null outcomes. In 13 LV sections of the parent feeder, ECF beats equal-total reduction in every section and beats no feedback in 12; LV26 loses

3.7 kWh against no feedback. Among nine representative published LV models, six have qualifying intervals and positive aggregate delivery differences, while three never qualify and remain unchanged. All reported section and representative-model trajectories pass their specified physical checks (Supplementary Sections S13.3–S13.4 and Supplementary Fig. 5). These comparisons show where the mechanism activates; related sections of one feeder are not independent feeder samples.

The aggregate gain is not shared equally. Among 418 participants in the primary feeder, 330 deliver more than without feedback over the 80 days and 88 deliver less; the largest individual loss is 7.3 kWh. The group-level replay establishes a feeder delivery opportunity, not a guarantee of gain for each connection (Supplementary Section S14.1).

3 Discussion

The scarce resource in request-based export sharing is usable network access, not merely the numerical size of an unused award. A large request can protect valuable export in one output realization and displace another participant's delivery in a different one. This explains why a charge tied only to the requester's shortfall, or a uniform cut in requests, can remove useful export together with idle permission.

ECF separates two decisions. The existing allocation rule determines today's awards; a completed-interval comparison determines whether some of today's unused access could have carried electricity from elsewhere. Only that finding informs a later request ceiling. Equal-total reductions and replay-free updates show that the result depends on both where and when the ceiling changes. In the public simulation, the added connection-point delivery is also largely visible as reduced source-terminal import. The evidence supports selective learning from completed operation without replacing the allocation rule.

Selection alone is insufficient for a reliable feedback loop. Available output can recover before a tight ceiling does, as the adverse unit-gain result shows. An uncertain post-event availability record can also change which intervals qualify: conservative bounds protect the decision in the tested overestimation cases but miss some recoverable delivery. The update qualifies an unused-award group, not individual responsibility within it. These features make delivered energy, ceiling recovery, record quality and participant-level effects relevant to an operating assessment.

The two studies test different parts of this argument. The controlled 141-bus study checks awards and delivered injections against nonlinear AC voltage, balance and two-ended apparent-power limits within declared synthetic branch ratings. The Australian simulations use measured household chronology and published network models, but construct requests and available export rather than observing issued limits and latent availability. Their physical screen covers customer voltage and native transformer limits. Gains vary with profile placement and output; a continuous 35-day block supplies most of the primary gain, and some participants lose delivery. These patterns locate the operating conditions under which the mechanism is valuable rather than supporting a uniform benefit claim.

An operational test would need linked requests, awards, metered delivery and a verified available-export record for the same connections and times. If access changes underlying availability through inverter control, storage or flexible demand, the replay must model that response. Such records would also allow the cost of measurement and computation to be compared with recovered export. Annual energy, market value and emissions are distinct questions for longer and more representative observations.

4 Methods

4.1 Requests, awards and available export

Let $\mathcal{N}$ be the participating export connections and $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ the distribution network. In interval t, participant i requests $r_i(t)$ before its available export $a_i(t)$ is known. Its fixed connection bound is C_i. Feedback changes a request ceiling $\psi_i(t) \in [0, C_i]$, initially C_i, so the admitted request is

$$r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}. \tag{1}$$

Allocation rule $\mathcal{R}$ then grants access $\mathbf{x}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \boldsymbol{\eta}_{\mathcal{R}}(t))$. Here $\mathcal{F}_{\mathcal{M}}$ represents network equations and limits under exogenous data $\mathbf{z}(t)$; $\boldsymbol{\eta}_{\mathcal{R}}$ contains rule parameters. Completed delivery is $y_i(t) = \min\{x_i(t), a_i(t)\}$, so $x_i - y_i$ is the unused award. Feedback modifies only the admitted-request input in a later interval, not the rule or network limits. These variables are interval-average powers; their sums over half-hours give the reported energy totals. Supplementary Table 1 collects the corresponding quantities.

The coordinator does not observe participants' private output signals or conditional distributions before allocation. It observes completed delivery and uses a post-event available-export record $a_i^{\mathrm{tel}}(t)$ for the replay. The controlled experiment supplies exact simulated availability. The Australian simulation constructs a profile-derived proxy and uses it as the primary replay record; neither this proxy nor modeled delivery is a field measurement of curtailed export.

Assumption 1 (Post-event available-export information)**.** *The comparison uses an available-export record a_i^{tel} for the completed interval. Its origin and time alignment must support the delivery calculation under an alternative award. In the controlled study the record is exact simulated availability; in the Australian study the primary record is the profile-derived proxy a_i^{prof}. Record sensitivities alter that proxy only for the replay decision. Neither record changes an award already issued.*

Assumption 2 (Same-state replay)**.** *Only admitted requests change in the completed-interval comparison. Topology, exogenous network data, participants, rule parameters and underlying available export remain fixed. Awards and power flows are recalculated. If a different award would change availability through inverter controls, storage, demand or portfolio dispatch, those responses must instead enter the replay model.*

4.2 Request choice and delivery displacement

The controlled study links a participant's request to later delivery through one conditional distribution of available export. Participant i observes a private signal s_i, forms a point forecast a_i^{fcst}, and assigns conditional cumulative distribution $F_i(\cdot \mid s_i)$ to its available export. Let $r_i^{\mathrm{ref}} = a_i^{\mathrm{fcst}}$ be its point-forecast request and $r_i^{\mathrm{chg}} = \gamma_i r_i^{\mathrm{ref}}$, $\gamma_i \geq 1$, a larger request. Holding other requests and the network state fixed, the participant values own delivery and faces a modeled shortfall cost $\vartheta \geq 0$ per unit of unused award, normalized by the value of delivered export:

$$U_i(\gamma_i; \vartheta) = \mathbb{E}\big[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\big]. \tag{2}$$

The selected multiplier is the smallest maximizer on the prespecified grid Γ. The same conditional available-export distribution determines this choice and the subsequent realization; the request and delivery curves are not generated independently. A static changed-request case is classified as a request–access–delivery externality if the changed group's expected payoff and unused award both

rise while aggregate expected delivery falls beyond the numerical tolerance. A productive high request raises the group's payoff without such a delivery loss. Dynamic cases use realized counterparts. This economic classification is distinct from ECF's operational replay test.

Within a smooth allocation regime, write $J_{ji} = \partial x_j / \partial r_i$ and $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$ for expected total delivery. Here $F_j(x_j)$ abbreviates the conditional cumulative distribution $F_j(x_j \mid s_j)$. Then

$$\begin{aligned} \frac{\partial U_i}{\partial r_i} &= J_{ii}[1 - (1+\vartheta)F_i(x_i)], \\ \frac{\partial \bar{Y}}{\partial r_i} &= \sum_j J_{ji}[1 - F_j(x_j)]. \end{aligned} \tag{3}$$

The first derivative need not have the sign of the second because the private payoff excludes delivery lost elsewhere. Where a request change crosses a bound or active network constraint, all reported classifications instead use matched finite differences. When both quantities are positive, outside-delivery loss divided by the additional unused award gives the diagnostic ratio plotted in Fig. 3(d). Its finite-change definition and relation to the local marginal shortfall cost are in Supplementary Section S2; that section also defines the capacity-normalized Jain index.

4.3 Allocation rules and physical checks

Five rules use the same admitted requests and controlled-study network domain (Supplementary Table 2). Weighted proportional (WP) scales all requests by one feasible factor. Equal reduction (ER) subtracts a common amount; equal access (EA) applies a common cap. Maximum export (ME) maximizes total authorized access. Fairness-aware allocation (FA) trades authorized access against a request-normalized equality term. Their one-parameter formulas, conic objectives, deterministic tie-breaks and sensitivity derivations are specified in Supplementary Sections S5–S8. Section S4 derives the matched request-reduction comparison. All rules allocate against an affine second-order-cone approximation of voltage and two-ended branch limits; the approximation is not treated as the final AC feasibility check.

The controlled medium-voltage case uses the published radial 141-bus network[37], a 0.70 demand scale and 30 fixed connection points. Synthetic, frozen branch ratings define this mechanism experiment; they are not native equipment nameplates. Q0 denotes zero participant reactive power and Q95 denotes 0.95 power factor with reactive-power absorption. Each matched reference, changed-request, award, observed-delivery and replay-delivery state is checked by nonlinear radial AC power flow at its fixed reactive-power setting. The check requires convergence, active- and reactive-power balance, $0.95 \le |V_b| \le 1.05$ pu and

$$\frac{\max_{e \in \{\mathrm{fr},\mathrm{to}\}} |S_{\ell,e}|}{\bar{S}_\ell^{\mathrm{syn}}} \le 1 \quad \text{for every monitored branch } \ell. \tag{4}$$

The branch test uses MVA at both ends, independently of rooted flow orientation. Supplementary Section S8 gives the frozen rating policy, calibration, and AC verification details.

4.4 Completed-interval replay and feedback

With unused-award tolerance $\epsilon_x > 0$, let $\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}$. After completion, ECF replaces only this group's admitted requests by the smaller of admission and metered delivery:

$$r_i^{\mathrm{rpl}}(t) = \begin{cases} \min\{r_i^{\mathrm{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r_i^{\mathrm{adm}}(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{5}$$

The same allocation rule and network data give replay access $\mathbf{x}^{\mathrm{rpl}}$. Available export then bounds replay delivery:

$$y_i^{\mathrm{rpl}} = \min\{x_i^{\mathrm{rpl}}, a_i^{\mathrm{tel}}\}.$$

The completed award and delivery are never overridden. The test qualifies the episode only if the replay releases access outside the unused-award group, raises delivery for at least one outside participant, raises aggregate delivery, and passes the applicable nonlinear AC checks. Each positive change must exceed τ_{rpl}. The test qualifies a set-level opportunity; it does not attribute a loss to each member individually. With a validated record-error bound $|a_i^{\mathrm{tel}} - a_i| \le \delta_i$, the conservative replay uses

$$\max\{y_i, a_i^{\mathrm{tel}} - \delta_i\}$$

as a lower bound on available export. Supplementary Section S3 proves the resulting lower bound on recovered delivery. Physical checks remain separate. With exact availability and the same recovery threshold, positive aggregate delivery recovery implies positive outside delivery and access recovery. We retain all three quantities to show how recovery occurs and to evaluate perturbed records; they are not three independent filters in the exact-record model.

Let $g(t) = 1$ when all replay conditions hold and zero otherwise. The target and next-interval ceiling are

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases}$$
$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad 0 \le \beta_{\mathrm{ECF}} \le 1. \tag{6}$$

Thus only a qualifying episode draws its unused-award group's ceilings toward delivery. Otherwise ceilings recover toward connection bounds. Consecutive nonqualifying intervals give geometric recovery for positive gain; a larger raw request cannot bypass a binding ceiling.

The shortfall-cost benchmark changes request choice but has no feedback state. Mismatch-only feedback updates a smoothed delivery-to-award ratio without replay. A second ablation applies ECF's update after every unused award without qualification. To assess whether update frequency alone explains ECF's advantage, a diagnostic control selects unused-award intervals at a fixed probability. The probability is 483/3091, the number of ECF qualifications divided by unused-award intervals in the original Australian evaluation. A hash of a fixed seed, date and half-hour selects intervals without consulting the replay or availability record beyond the completed delivery needed to determine the unused-award group. We report all 12 fixed schedules and average them within each date. This comparison was specified after the original results were known; it is a diagnostic matched in expected update frequency, not an independent validation. Every schedule follows its own awards, delivery and ceiling state.

The equal-total control matches ECF's total request reduction in each interval but spreads it proportionally across connections. If $r_i^{\mathrm{cap}} = \min\{r_i, C_i\}$ and $\Delta R_{\mathrm{ECF}} = \sum_i (r_i^{\mathrm{cap}} - r_i^{\mathrm{adm,ECF}})$, the control

admits

$$r_i^{\mathrm{adm,uni}} = r_i^{\mathrm{cap}} \left(1 - \frac{\Delta R_{\mathrm{ECF}}}{\sum_j r_j^{\mathrm{cap}}}\right) \tag{7}$$

when the denominator is positive, and retains the capacity-limited request otherwise. This isolates where requests are reduced. With total authorization $X = \sum_i x_i$ and unused authorization $H = \sum_i (x_i - y_i)$, the accounting identity $Y = X - H$ yields

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = (X^{\mathrm{ECF}} - X^{\mathrm{uni}}) + (H^{\mathrm{uni}} - H^{\mathrm{ECF}}). \tag{8}$$

It does not assert that either term has a fixed sign. Supplementary Section S4 derives the corresponding redistribution conditions.

4.5 Controlled experiments

A five-node conditional available-export tree uses standardized support $(-2, -1, 0, 1, 2)$ with probabilities $(1, 4, 6, 4, 1)/16$ and scale σ. The participant's forecast anchors both its request and the later node realization. The static grid spans 30 locations, five rules, Q0 and Q95, five uncertainty levels and nine shortfall-cost levels, giving 13,500 request-choice cases. The selected payoff-maximizing multiplier is held fixed within a 16-interval trajectory. Three trajectories have identical node counts but differ in ordering: clustered, intermittent and shock–recovery. Changed requests come from one participant, five electrically nearby participants, or five dispersed participants. In the five-participant cases, each uses its own best-response multiplier and all five requests change simultaneously. The matched design comprises 1,620 trajectories and 25,920 interval comparisons across the controls and gains. Supplementary Table 3 gives every factor, numerical tolerance and parameter level. The post-event record is exact simulated availability in this controlled study.

4.6 Australian public-data simulation

The external simulation combines a published unbalanced 11-kV feeder with 1,255 service connections and its LV sections[38] with half-hour PV and consumption profiles from 300 Ausgrid solar homes[39]. These sources have no common customer identifier. The primary deterministic profile-to-service mapping is fixed before assessment, and separate placements test its sensitivity. The forward-uptake scenario assigns 418 connections export participation and scales PV to an approximately 9.9-kWp mean[40]. It is not a historical reconstruction of issued operating envelopes. The day-to-day forecast increment is estimated from 2010–2011 data; 2011–2012 data select fixed feeder settings; 2012–2013 provide the assessment chronology. A day is eligible only when mapped profiles are complete on that day and the previous day. Excluding missing and daylight-saving-transition records leaves 80 network days, or 3,840 half-hours per control.

For mapped profile $\iota(i)$, scaled gross PV generation GG, general consumption GC and controlled load CL yield

$$a_i^{\mathrm{prof}}(d, \tau) = \left[\lambda_{\mathrm{PV}} \mathrm{GG}_{\iota(i)}(d, \tau) - \lambda_L (\mathrm{GC}_{\iota(i)}(d, \tau) + \mathrm{CL}_{\iota(i)}(d, \tau))\right]^+ . \tag{9}$$

Modeled delivery under control c is $y_i^c = \min\{x_i^c, a_i^{\mathrm{prof}}\}$; the unperturbed profile proxy also supplies the primary replay record. Raw requests use the previous day's same-half-hour proxy plus a positive 90th-percentile increment from same-half-hour day-to-day changes. Contemporaneous household

demand is fixed as an exogenous input to this retrospective allocation; load-forecast error is not modeled.

The primary feeder settings use a 0.30 load multiplier and a 1.07-pu allocation-stage upper-voltage limit, below the 1.10-pu physical limit. For each half-hour, a zero-anchored feasible-factor search scales admitted requests under nonlinear unbalanced AC power flow. An independent factor-grid check of all 19,200 primary allocation calls found no feasible sampled factor above the returned one. The sampled check supports these allocations but does not prove global maximality between grid points (Supplementary Section S10.4). Load-only, authorized-export and modeled-delivery states are screened separately for convergence, customer voltage and native transformer ratings. Because the public model lacks usable native line ampacities, line-thermal certification is outside this physical screen. The replay threshold is 2 W, a numerical decision threshold rather than an instrument-accuracy claim.

Each primary day begins with ceilings at C_i and compares no feedback, previous-day persistence, ECF at $\beta_{\mathrm{ECF}} = 0.5$, and equal-total reduction. The additional comparisons vary gain, replay qualification, cross-day memory, record perturbations and error bounds, profile placement, PV capacity, participant adaptation and published LV networks while retaining the documented fixed inputs for each matched comparison. Source-terminal power from the same AC solves accounts for upstream exchange. The request quantile and fixed feeder settings are calculated from the earlier source years. Earlier evaluations had already been inspected when the present scenario was constructed, so its 80-day comparison is a fixed-scenario reevaluation rather than a new blind validation. Supplementary Section S10.3 records the selection sequence and full model construction.

4.7 Statistical analysis

The primary comparison unit in the Australian study is one paired network day. A day contains 48 half-hours on the same network and profile mapping; participants, half-hours and nested LV sections are not treated as independent network samples. Energy differences are computed within each day before averaging. We report ordinary paired-day percentile intervals and a date-dependent wild-bootstrap interval that allows short-range correlation between nearby eligible dates. Both use 10,000 replicates. The request- adaptation experiment averages three seeds within each date; the frequency-calibrated-update comparison averages 12 schedules within each date. Intervals are pointwise and do not adjust for multiple comparisons. Supplementary Section S10.3 gives the calendar covariance, exact eligibility rules and full numerical settings.

Author contributions

Junjie Huang: conceptualization, methodology, software, formal analysis, investigation, visualization, data curation, and writing—original draft. Tao Yu and Yusheng Xue: conceptualization, supervision, funding acquisition, and writing—review and editing. Zhenning Pan, Yufeng Wu, and Wencong Xiao: methodology, validation, and writing—review and editing. Rankai Zhu: validation, data curation, and writing—review and editing. Jie Huang: validation and writing—review and editing.

Competing interests

The authors declare no competing interests.

Data availability

Calculated results, figure source data, input-source identifiers, preprocessing instructions, date selections and network-profile mapping tables are available in the public project repository, `https://github.com/MarxHuang/ecf-export-access`. The source 141-bus network, CSIRO/GridQube network collection (doi:10.25919/ghnz-bk28) and Ausgrid Solar Home Electricity Data remain with their original distributors under their respective terms; raw third-party records are not redistributed. The repository distinguishes deposited calculated data from those external inputs required for a full simulation rerun.

Code availability

Simulation, preprocessing and analysis code, parameter files, environment specifications and reproduction instructions are available at `https://github.com/MarxHuang/ecf-export-access`. The repository identifies its deposited revision and provides checks of saved results separately from complete reruns, which require the source datasets described above.

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